

Summary so far

Data preparation

Raw input is OptionMetrics SPX option quotes plus the IvyDB Forward_Price table. For each trading day we:

1. Filter quotes to OTM options plus a tight ATM band ($|k| < 0.01$ with $k = \log(K/F)$, where F is the same-expiry forward).
2. Drop quotes with vega < 0.5 , $\sigma_{IV} \notin [0.01, 3.0]$, or zero traded volume.
3. Fit a vega-weighted Nadaraya–Watson kernel smoother in $(\log \tau, k)$ space, using the OptionMetrics manual form

$$\hat{\sigma}(\tau_j, k_j) = \frac{\sum_i V_i \sigma_i \phi(x_{ij}, y_{ij})}{\sum_i V_i \phi(x_{ij}, y_{ij})}, \quad \phi(x, y) = \exp\left(-\frac{x^2}{2h_\tau} - \frac{y^2}{2h_m}\right),$$

with $x_{ij} = \log(\tau_i/\tau_j)$, $y_{ij} = k_i - k_j$, and bandwidths (variances) $h_\tau = 5.00 \times 10^{-2}$, $h_m = 1.00 \times 10^{-3}$.

4. Evaluate $\hat{\sigma}$ on a fixed 20×20 grid: k uniform over $[-0.10, 0.10]$ and τ geometric over $[0.04, 1.00]$ years.

Output is one row per date with 400 flattened columns `iv_{moneyness}_{tau}` (column order: τ outer, k inner). Sparse-data guards (effective sample size > 2 , no single observation holding > 0.85 of the kernel mass) blank out cells where the smoother would be unstable.

Task

The current evaluation task is **level** / **precovid** / **c63** / **h21**:

- **level**: predict raw IV σ , not its log-difference.
- **precovid**: restrict the panel to dates $\leq 2019-12-31$ to avoid the COVID-era regime break (so the model is judged on a stationary regime).
- **c63**: context length $L = 63$ trading days (≈ 3 months).
- **h21**: forecast horizon $H = 21$ trading days (≈ 1 month).

This gives $T = 2,768$ daily surfaces. The standard 70/10/20 split puts the train period ending 2016-09-12, with 533 rolling test windows spanning 2017-10-19 $\rightarrow$ 2019-12-02. All metrics below are computed on this test set, in original-IV units (vol points).

Best models (tuning winners)

For every architecture we ran a grid search and kept the combo with the lowest validation loss (validation = the 10% slice between train and test). VAR is not tuned — its lag p is picked by an information criterion on the training set.

- **VAR(BIC)** — Vector autoregression on the 400-dim flattened surface, fit by OLS with lag p chosen to minimise BIC over $p \in \{1, \dots, 5\}$. The BIC criterion selects $p = 1$ (so the model is effectively a 400×400 one-step transition matrix iterated forward). Strong, almost embarrassingly simple baseline that exploits the high cross-sectional and temporal correlations of the IV surface. Parameters: 1.60×10^5 ($K^2 p + K$ with $K = 400$, $p = 1$).
- **DLinear** — Purely linear time-series baseline that decomposes each channel into a moving-average trend and a residual, then maps each via a single dense layer over the context window. Winner: `kernel_size = 13`, `lr =  $4.00 \times 10^{-3}$` , `weight decay = 0`, MSE loss (val = 4.05×10^{-2}). Parameters: 1.08×10^6 .
- **HOT(product)** — Tensorised transformer over the 20×20 surface. Patches the time axis, then applies multi-head attention with a Kronecker *product* factorisation across the moneyness and τ axes (a tensor attention that respects grid structure). Winner: `d_hidden = 64`, `d_mlp = 32`, `n_blocks = 2`, `n_head = 4`, `patch = 3`, RoPE positional embedding, no spatial PE, `lr =  $1.00 \times 10^{-3}$` , `weight decay =  $1.00 \times 10^{-3}$` (val = 3.29×10^{-2}). Parameters: 8.14×10^4 .

- **HOT(sum)** — Same architecture as HOT(product) but with the Kronecker *sum* factorisation (separately attends along moneyness and τ , then adds). Winner: $d_{\text{hidden}} = 64, d_{\text{mlp}} = 16, n_{\text{blocks}} = 4, n_{\text{head}} = 2, \text{patch} = 9, \text{lr} = 1.00 \times 10^{-3}, \text{weight decay} = 1.00 \times 10^{-3}$ (val = 3.60×10^{-2}). Parameters: 1.49×10^5 .
- **DynGWN** — Graph WaveNet variant where the 400 surface cells form a graph that combines a fixed grid-adjacency component with a learned adaptive component. Winner (graph_mode = grid_plus_adaptive): $n_{\text{hid}} = 8, \text{blocks} = 2, \text{layers} = 2, \text{kernel} = 2, \text{dropout} = 0.3, \text{lr} = 1.00 \times 10^{-2}, \text{weight decay} = 1.00 \times 10^{-3}$ (val = 3.18×10^{-2}). Parameters: 3.67×10^4 .
- **PatchTST** — Channel-independent transformer over time-axis patches, shared across the 400 cells. Winner: $\text{patch_len} = 3, \text{stride} = 3, d_{\text{model}} = 32, n_{\text{heads}} = 2, n_{\text{layers}} = 1, d_{\text{ff}} = 128, \text{dropout} = 0.3, \text{RevIN off}, \text{lr} = 3.00 \times 10^{-4}, \text{weight decay} = 1.00 \times 10^{-4}$ (val = 3.12×10^{-2}). Parameters: 2.83×10^4 .

Metrics

Standard pointwise metrics: MSE, RMSE, MAE, RSE, bias, IC. IC is the per-window Spearman rank correlation between predicted and true IV across the 400 cells, then averaged over windows and horizons.

Surface-shape metrics (LSTS components, all squared and averaged over the test windows, horizons, and the relevant axis):

$$\mathcal{L}_{\text{level}} = \mathbb{E} \left[\left(\bar{\hat{\sigma}} - \bar{\sigma} \right)^2 \right]$$

$$\mathcal{L}_{\text{skew}} = \mathbb{E} \left[\left(\Delta_k \hat{\sigma} - \Delta_k \sigma \right)^2 \right]$$

$$\mathcal{L}_{\text{term}} = \mathbb{E} \left[\left(\Delta_\tau \hat{\sigma} - \Delta_\tau \sigma \right)^2 \right]$$

$$\mathcal{L}_{\text{curv}} = \mathbb{E} \left[\left(\Delta_k^2 \hat{\sigma} - \Delta_k^2 \sigma \right)^2 \right]$$

with Δ_k a finite difference along moneyness divided by the local grid spacing, Δ_τ likewise along τ , and Δ_k^2 the uniform-grid second difference along moneyness. The overbar denotes the mean across the 20×20 surface.

Persist(ref) is the naive last-value baseline: $\hat{\sigma}_{t+h} = \sigma_t$ for every horizon h .

In every table that follows, **bold** marks the best value in the row and underline marks the second best. For bias the ranking is by absolute value (closest to zero is best); for IC mean and per-horizon IC, larger (more positive) is best; for every other metric, smaller is best.

Overall test-set results

Metric	Persist(ref)	VAR(1)	DLinear	HOT(product)	HOT(sum)	DynGWN	PatchTST
MSE	7.29×10^{-4}	6.13×10^{-4}	<u>6.33×10^{-4}</u>	9.73×10^{-4}	1.07×10^{-3}	8.50×10^{-4}	6.92×10^{-4}
RMSE	2.70×10^{-2}	2.48×10^{-2}	<u>2.52×10^{-2}</u>	3.12×10^{-2}	3.27×10^{-2}	2.92×10^{-2}	2.63×10^{-2}
MAE	<u>1.72×10^{-2}</u>	1.85×10^{-2}	1.70×10^{-2}	2.16×10^{-2}	2.18×10^{-2}	1.91×10^{-2}	1.72×10^{-2}
RSE	6.68×10^{-1}	6.13×10^{-1}	<u>6.23×10^{-1}</u>	7.72×10^{-1}	8.09×10^{-1}	7.22×10^{-1}	6.51×10^{-1}
Bias	<u>-7.19×10^{-4}</u>	$+6.56 \times 10^{-3}$	$+3.89 \times 10^{-4}$	-7.98×10^{-3}	-1.48×10^{-2}	-6.67×10^{-3}	-4.15×10^{-3}
IC mean	$+9.49 \times 10^{-1}$	$+9.63 \times 10^{-1}$	$+9.55 \times 10^{-1}$	$+9.46 \times 10^{-1}$	$+9.50 \times 10^{-1}$	$+9.52 \times 10^{-1}$	<u>$+9.55 \times 10^{-1}$</u>
IC std	8.91×10^{-2}	7.02×10^{-2}	<u>7.29×10^{-2}</u>	7.99×10^{-2}	8.43×10^{-2}	8.41×10^{-2}	<u>8.24×10^{-2}</u>
$\mathcal{L}_{\text{level}}$	6.12×10^{-4}	5.31×10^{-4}	<u>5.37×10^{-4}</u>	8.57×10^{-4}	9.47×10^{-4}	7.45×10^{-4}	5.96×10^{-4}
$\mathcal{L}_{\text{skew}}$	1.12×10^{-2}	8.94×10^{-3}	1.33×10^{-2}	1.79×10^{-2}	2.14×10^{-2}	1.76×10^{-2}	<u>9.96×10^{-3}</u>
$\mathcal{L}_{\text{term}}$	2.86×10^{-2}	1.80×10^{-2}	2.46×10^{-2}	<u>2.01×10^{-2}</u>	2.14×10^{-2}	2.02×10^{-2}	2.11×10^{-2}
$\mathcal{L}_{\text{curv}}$	8.23×10^{-8}	6.14×10^{-8}	1.62×10^{-7}	<u>2.20×10^{-7}</u>	3.44×10^{-7}	4.69×10^{-7}	<u>6.27×10^{-8}</u>

Per-horizon results

Spearman IC

h	Persist(ref)	VAR(1)	DLinear	HOT(product)	HOT(sum)	DynGWN	PatchTST
1	$+9.90 \times 10^{-1}$	<u>$+9.90 \times 10^{-1}$</u>	$+9.87 \times 10^{-1}$	$+9.59 \times 10^{-1}$	$+9.69 \times 10^{-1}$	$+9.86 \times 10^{-1}$	$+9.89 \times 10^{-1}$
5	$+9.66 \times 10^{-1}$	$+9.70 \times 10^{-1}$	$+9.64 \times 10^{-1}$	$+9.52 \times 10^{-1}$	$+9.59 \times 10^{-1}$	$+9.66 \times 10^{-1}$	<u>$+9.67 \times 10^{-1}$</u>
10	$+9.47 \times 10^{-1}$	$+9.60 \times 10^{-1}$	$+9.49 \times 10^{-1}$	$+9.47 \times 10^{-1}$	$+9.50 \times 10^{-1}$	$+9.51 \times 10^{-1}$	<u>$+9.54 \times 10^{-1}$</u>
21	$+9.25 \times 10^{-1}$	$+9.52 \times 10^{-1}$	<u>$+9.41 \times 10^{-1}$</u>	$+9.36 \times 10^{-1}$	$+9.39 \times 10^{-1}$	$+9.30 \times 10^{-1}$	$+9.36 \times 10^{-1}$

MSE

h	Persist(ref)	VAR(1)	DLinear	HOT(product)	HOT(sum)	DynGWN	PatchTST
1	1.28×10^{-4}	<u>1.30×10^{-4}</u>	1.41×10^{-4}	4.23×10^{-4}	4.70×10^{-4}	2.08×10^{-4}	1.80×10^{-4}
5	4.56×10^{-4}	4.30×10^{-4}	<u>4.34×10^{-4}</u>	6.99×10^{-4}	7.94×10^{-4}	5.27×10^{-4}	4.61×10^{-4}
10	7.38×10^{-4}	6.39×10^{-4}	<u>6.43×10^{-4}</u>	9.52×10^{-4}	1.06×10^{-3}	8.23×10^{-4}	6.72×10^{-4}
21	1.17×10^{-3}	8.75×10^{-4}	<u>9.40×10^{-4}</u>	1.40×10^{-3}	1.44×10^{-3}	1.39×10^{-3}	1.08×10^{-3}

$\mathcal{L}_{\text{level}}$

h	Persist(ref)	VAR(1)	DLinear	HOT(product)	HOT(sum)	DynGWN	PatchTST
1	1.01×10^{-4}	<u>1.03×10^{-4}</u>	1.09×10^{-4}	3.24×10^{-4}	3.77×10^{-4}	1.63×10^{-4}	1.51×10^{-4}
5	3.69×10^{-4}	<u>3.60×10^{-4}</u>	3.53×10^{-4}	5.92×10^{-4}	6.87×10^{-4}	4.45×10^{-4}	3.84×10^{-4}
10	6.14×10^{-4}	<u>5.51×10^{-4}</u>	5.39×10^{-4}	8.37×10^{-4}	9.38×10^{-4}	7.17×10^{-4}	5.73×10^{-4}
21	9.98×10^{-4}	7.73×10^{-4}	<u>8.23×10^{-4}</u>	1.27×10^{-3}	1.30×10^{-3}	1.25×10^{-3}	9.44×10^{-4}

 $\mathcal{L}_{\text{skew}}$

h	Persist(ref)	VAR(1)	DLinear	HOT(product)	HOT(sum)	DynGWN	PatchTST
1	<u>3.76×10^{-3}</u>	3.46×10^{-3}	7.44×10^{-3}	1.72×10^{-2}	1.76×10^{-2}	1.03×10^{-2}	4.10×10^{-3}
5	8.28×10^{-3}	7.26×10^{-3}	1.03×10^{-2}	1.61×10^{-2}	1.85×10^{-2}	1.21×10^{-2}	<u>7.45×10^{-3}</u>
10	1.16×10^{-2}	9.44×10^{-3}	1.48×10^{-2}	1.71×10^{-2}	2.14×10^{-2}	1.65×10^{-2}	<u>9.91×10^{-3}</u>
21	1.57×10^{-2}	1.13×10^{-2}	1.48×10^{-2}	2.01×10^{-2}	2.47×10^{-2}	2.62×10^{-2}	<u>1.36×10^{-2}</u>

 $\mathcal{L}_{\text{term}}$

h	Persist(ref)	VAR(1)	DLinear	HOT(product)	HOT(sum)	DynGWN	PatchTST
1	7.72×10^{-3}	8.90×10^{-3}	1.42×10^{-2}	1.87×10^{-2}	1.65×10^{-2}	1.01×10^{-2}	<u>7.85×10^{-3}</u>
5	2.34×10^{-2}	1.70×10^{-2}	2.55×10^{-2}	1.95×10^{-2}	1.97×10^{-2}	<u>1.80×10^{-2}</u>	1.92×10^{-2}
10	3.26×10^{-2}	1.92×10^{-2}	2.68×10^{-2}	<u>2.02×10^{-2}</u>	2.19×10^{-2}	2.14×10^{-2}	2.37×10^{-2}
21	3.39×10^{-2}	1.97×10^{-2}	2.37×10^{-2}	<u>2.07×10^{-2}</u>	2.32×10^{-2}	2.32×10^{-2}	2.39×10^{-2}

h	Persist(ref)	VAR(1)	DLinear	HOT(product)	HOT(sum)	DynGWN	PatchTST
1	5.28×10^{-8}	<u>4.34×10^{-8}</u>	2.66×10^{-7}	2.39×10^{-7}	3.07×10^{-7}	3.92×10^{-7}	4.20×10^{-8}
5	7.00×10^{-8}	<u>5.42×10^{-8}</u>	1.64×10^{-7}	2.08×10^{-7}	3.04×10^{-7}	3.36×10^{-7}	5.15×10^{-8}
10	8.27×10^{-8}	<u>6.26×10^{-8}</u>	1.30×10^{-7}	2.12×10^{-7}	3.43×10^{-7}	4.34×10^{-7}	6.22×10^{-8}
21	9.95×10^{-8}	7.12×10^{-8}	1.22×10^{-7}	2.33×10^{-7}	3.73×10^{-7}	5.90×10^{-7}	<u>7.80×10^{-8}</u>